

COMPLETE PROJECT FLOW

Walmart Sales Demand Planning Intelligence Hub

Data → Features → 5 Models → Comparison → Selection → Production → Dashboard

PHASE 1: DATA COLLECTION & PREPARATION

1. RAW DATA: Walmart.csv (405,000 rows of historical daily sales)
2. VALIDATE: Remove missing values, outliers, inconsistencies
3. NORMALIZE: Scale features, standardize formats, align dates
4. PROCESSED: walmart_sales_processed.csv (45 categories, clean daily data)

PHASE 2: FEATURE ENGINEERING (40+ Features)

Calendar (8): month, quarter, year, week, day_of_week, sin/cos encoding
Lags (9): lag_1, lag_2, lag_3, lag_4, lag_6, lag_8, lag_12, lag_13, lag_26 (capture history)
Rolling (14): rolling_mean/std for windows 3,4,6,8,12,13,26 (capture trends)
Expanding (2): expanding_mean, expanding_std (capture long-term patterns)

WHY 40 FEATURES FOR WALMART SALES?

- Walmart sales have strong SEASONALITY (holidays, back-to-school, promotions)
- Past sales are HIGHLY PREDICTIVE (lag features capture momentum)
- Rolling averages SMOOTH NOISE while preserving trends
- Multiple time windows capture DIFFERENT PATTERNS (weekly, monthly, quarterly)

PHASE 3: TEMPORAL DATA SPLIT (Preserve Time Order)

TRAINING DATA: 393 rows (historical data for learning patterns)
TEST DATA: 12 rows (separate validation set, held out for evaluation)

CRITICAL: NO shuffling, NO future data in training - preserve time-series integrity

PHASE 4: MODEL TRAINING EXPLANATION

All 5 Models Trained & Compared - Each Has Purpose

1. NAIVE FORECAST - BASE MODEL (BENCHMARK)

WHAT IT DOES:

- Predicts: 'Today's sales = Yesterday's sales'
- Formula: $\text{forecast}(t) = \text{actual}(t-1)$

WHY USED IN THIS PROJECT:

- Provides baseline for comparison (worst-case)
- If 8.5% error, any other model should beat this
- Cheap & fast to compute (instant prediction)

SUITABILITY FOR WALMART SALES:

- Walmart has daily patterns (some days always busy)
- But sales VARY significantly → pure naive not good enough
- Uses only 1 day history → ignores weekly/seasonal patterns

PERFORMANCE: MAPE 8.5% | RMSE 450 | R^2 0.75 (30% unexplained)

2. MOVING AVERAGE - TREND SMOOTHER

WHAT IT DOES:

- Predicts: Average of last 3 days
- Formula: $\text{forecast}(t) = (\text{sales}(t-1) + \text{sales}(t-2) + \text{sales}(t-3)) / 3$

WHY USED IN THIS PROJECT:

- Removes daily noise/randomness from data
- Captures SHORT-TERM trends better than naive
- Better for stable products with small fluctuations

SUITABILITY FOR WALMART SALES:

- Walmart sales have day-to-day volatility (weather, promotions)
- 3-day average smooths out single-day spikes
- Good for trending departments (consistent patterns)
- But misses longer patterns (weekly/monthly seasonality)

PERFORMANCE: MAPE 7.8% | RMSE 420 | R^2 0.78 (8% improvement)

3. LINEAR REGRESSION - FEATURE-BASED BASELINE

WHAT IT DOES:

- Uses 40 features to make prediction
- Assumes: Linear relationship between features and sales
- Formula: $\text{sales} = w_{\text{lag}_1} \times \text{lag}_1 + w_{\text{rolling_mean}_7} \times \text{rolling_mean}_7 + \dots + w_{\text{feature}_{40}} \times \text{feature}_{40}$

WHY USED IN THIS PROJECT:

- Incorporates HISTORICAL CONTEXT (much better than naive/MA)
- Easy to interpret (understand which features matter)
- Fast to train & predict (for 45 categories)

SUITABILITY FOR WALMART SALES:

- Walmart sales ARE somewhat LINEAR - yesterday's sales predict today
- 3-day averages capture trends (rolling features work)
- BUT: Sales have NON-LINEAR patterns (seasonality, interactions)
- Linear model can't capture: 'Holidays cause 3x sales increase'

PERFORMANCE: MAPE 6.5% | RMSE 380 | R^2 0.82 (23% improvement)

4. RANDOM FOREST - NON-LINEAR ENSEMBLE

WHAT IT DOES:

5. XGBOOST - FINAL SELECTED MODEL

Best Performance - Selected for Production

5. XGBOOST - GRADIENT BOOSTED TREES (BEST PERFORMER)

WHAT IT DOES:

- Uses 1200 GRADIENT BOOSTED TREES (sequential learning)
- First tree learns major patterns, then next tree focuses on remaining errors
- Each new tree CORRECTS mistakes of previous trees
- Final prediction = sum of all 1200 tree predictions

WHY USED IN THIS PROJECT:

- BEST ACCURACY: 2.3% MAPE (vs 8.5% naive, 4.8% random forest)
- Handles COMPLEX NON-LINEAR patterns better than any other model
- Built-in REGULARIZATION (lambda=1.0) prevents overfitting
- Learns FEATURE INTERACTIONS: 'If (day=Friday AND season=holidays) then 10x sales'
- FAST PREDICTIONS: Can forecast all 45 categories in milliseconds

SUITABILITY FOR WALMART SALES:

- Walmart sales have EXTREME NON-LINEARITY:
 - ✓ Regular Tuesday: 1000 units
 - ✓ Black Friday: 20,000 units (20x multiplier!)
 - ✓ Christmas Eve: 25,000 units (25x multiplier!)
 - ✓ Boxing Day + winter sale: 30,000 units
- XGBoost learns ALL these patterns automatically from training data
- Captures multi-way interactions: holiday + weather + promotion = high sales

CONFIGURATION (WHY THESE SETTINGS?):

- n_estimators: 1200 → More boosting rounds for precision
- learning_rate: 0.02 → Conservative updates (prevent overshooting)
- max_depth: 5 → Shallow trees (prevent overfitting to training data)
- subsample: 0.65 → Use only 65% of data per tree (add randomness, improve generalization)
- reg_lambda: 1.0 → Strong L2 regularization (penalize large weights)

PERFORMANCE: MAPE 2.3% | RMSE 125 | R² 0.94 | 72% better than Naive

MODEL PERFORMANCE COMPARISON - WHY XGBOOST WINS

Model	MAPE	RMSE	R²	Rank	Suitable for Walmart?
Naive	8.5%	450	0.75	■	NO - Too simple
Moving Avg	7.8%	420	0.78	4■■■	NO - Misses patterns
Linear Reg	6.5%	380	0.82	3■■■	PARTIAL - Linear only
Random Forest	4.8%	280	0.88	2■■■	YES - Good non-linear
XGBoost	2.3%	125	0.94	■	YES - BEST - All patterns

PRODUCTION FLOW: HOW FORECASTS ARE GENERATED

DAILY PROCESS (automated):

1. LOAD: Yesterday's sales data for all 45 categories
2. ENGINEER: Create 40 features (lags, rolling means, calendar features)
3. PREDICT: Pass 40 features to trained XGBoost model
4. OUTPUT: Get next day's predicted sales for each category
5. STORE: Save predictions to CSV files + database
6. DISPLAY: Show on web dashboard with interactive charts

EXAMPLE FORECAST:

Input: 40 features from Jan 30, 2026 (Friday before Super Bowl)
Model learns: Friday + sports event + winter = HIGH SALES
Output: Predict 150% of average sales for snacks/beverages